

THE USE OF ACCELERATED AGEING PROCEDURES TO PREDICT
THE LONG TERM STRENGTH OF GRC COMPOSITES

K. L. LITHERLAND, D. R. OAKLEY, B. A. PROCTOR
Pilkington Brothers Ltd., R&D Laboratories, Lathom,
Nr. Ormskirk, Lancashire, United Kingdom.

(Communicated by A.J. Majumdar)
(Received Feb. 25, 1981)

ABSTRACT

A method has been developed to measure the direct tensile strength of glassfibre strands in a cement environment. Changes in the strength of strands of Cem-FIL A.R. glassfibre in Portland Cement have been accelerated by ageing in hot water at several temperatures and the temperature dependence determined. Similar accelerated ageing has been carried out with GRC composites and the results compared with strength changes in material exposed to natural weathering in a variety of climates. Predictions are given for expected strengths of Cem-FIL GRC composites over a hundred (or more) years.

Es wurde eine Methode zur Messung der unmittelbaren Zugfestigkeit von ungezwirnten Glasseidebündeln in Zement entwickelt. Änderung in der Festigkeit ungezwirnter Bündel von Cem-FIL A.R.-Glasseide in Portland-Zement wurde durch Alterung in heissem Wasser bei mehreren Temperaturen beschleunigt, und es wurde die Temperaturabhängigkeit bestimmt. Ähnliche beschleunigte Alterung wurde bei GRC-Verbundstoffen durchgeführt, und die Ergebnisse wurden mit Festigkeitsänderungen bei natürlicher Bewitterung unter verschiedenen klimatischen Bedingungen ausgesetzten Materialien verglichen. Langzeit-Festigkeitswerte von Cem-FIL GRC-Verbundstoffen werden für hundert Jahre (oder mehr) vorhergesagt.

Introduction

The introduction of any new material into use in modern industry requires a knowledge of long term strength retention under working conditions. This may be obtained from exposure of test samples to "real" working conditions over the expected working lifetimes - or estimated from the behaviour of samples exposed to much more aggressive accelerated ageing conditions over shorter time periods.

In the Building and Construction Industries the former approach is initially of limited practical value in the assessment of new materials since expected working lifetimes often extend over many tens of years. However it is still essential in providing, ultimately, the only unequivocal long term strength data and this point was recognised in the very early days of the development of alkali resistant (AR) glass fibres for cement reinforcement (1, 2, 3) by the establishment of "real weathering" test programmes for Glassfibre Reinforced Cement materials, (GRC). The results of these programmes (4, 5, 6) are already proving to be of great value.

Nonetheless, in supporting the commercialisation and market development of AR glass fibre for cement reinforcement, the need for a reliable way of estimating the very long term strength of GRC has become increasingly apparent. Clearly, to be useful in the relatively early stages of development these estimates had to be based on the second, accelerated ageing, approach since they were required to extend over many tens of years. Gradually, over the past 6 to 8 years, experimental information has been acquired and a methodology of accelerated ageing developed - often prompted by the queries of design engineers, potential GRC users, approving authorities or standards bodies (6, 7).

This paper now attempts to explain this accelerated test method, to make available the data on which predictions are based and to provide long term (e.g. 50 to 100 years or more) strength estimates for some standard and widely used forms of GRC (4, 6, 7, 8).

Essentially the long term strength predictions are based on the availability of three main bodies of experimental data.

- (i) knowledge of the direct tensile strength of strands of Cem-FIL AR glass fibre reinforcement in cement environments, after exposure to hot, wet, conditions.
- (ii) knowledge of the strengths of Cem-FIL GRC composites after accelerated ageing in similar aggressive conditions.

and, as a basis for comparison,

- (iii) knowledge of the strengths of Cem-FIL GRC composites after weathering for some years in a variety of real climates.

Strengths of Glassfibre Strands

Direct knowledge of the strength of the fibre reinforcement is an important factor in understanding the behaviour of any composite material, although the strength of the composite may in fact be influenced by a number of factors, including the strength of the matrix and the matrix/fibre interfacial bond, in addition to the fibre strength.

In the case of glass fibre reinforced cements and mortars the strength behaviour of the cement or mortar matrix was reasonably well understood and was predictable over a period of years. The strength of fibre/matrix bonds was less well known, but it was the strength behaviour of the A.R. glass fibre itself which was least understood - it had even been suggested (9) that the strength of the composite was not directly controlled by the strength of the fibres.

Against this background it was clearly important to be able to measure, directly, the tensile strength of glass fibre strands in a hardened and cured cement environment which simulated and represented the conditions in the composite as closely as possible.

The Strand-in-Cement test (SIC test) was devised for this purpose.

For this test a small block of cement paste or cement/sand mortar is cast around part of a typical, commercial, multifilament strand. The strand outside the cement is strengthened and protected by impregnation with resin and this is continued for a short distance into the cement to prevent damage and flexing at the edge of the block. Adhesion between resin and cement is prevented by small plasticine grommets and a final protection is given to the strand outside the cement by a coat of wax. Typical dimensions and construction of an SIC specimen are shown in Figure 1.

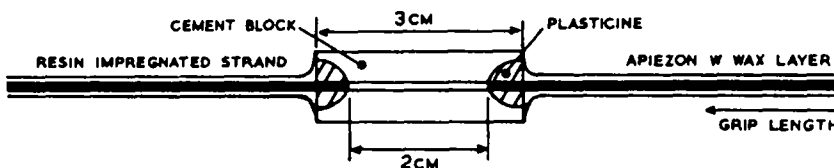


FIG.1 Strand-in-Cement (SIC) Specimen

After casting the cement is allowed to set and "cure" for 24 hours at 100% RH at room temperature. Specimens are then transferred to a suitable storage environment for the required period, removed and tested in direct tension - the strand ends being gripped in pneumatically operated grips. The tensile strength of the strand is then calculated from the breaking load, measured strand tex, and fibre density.

This specimen and test procedure has proved to be an extremely valuable and flexible experimental tool enabling comparisons to be made between glassfibre strength retention in different cement and mortar systems, between strengths in cement and in alkaline solutions, between fibre strength retention and chemical durability tests (3), and between fibre and composite strength changes.

For studies of fibre strength loss under accelerated ageing conditions several groups of ten specimens were immersed in water at various temperatures. A group of ten specimens was removed at intervals and the mean strength of these ten specimens taken as a measure of the glassfibre strength. The results of very many such accelerated ageing tests with Fibreglass Cem-FIL AR Glassfibre in Rapid Hardening Portland cement mortar blocks are shown in Figure 2 where each test point is the average strength of a group of ten individual SIC specimens.

The lines fitted through the data points at each test temperature in Figure 2 are based on an assumed $(\text{Strength})^{-1}$ versus $\log(\text{time})$ relationship. (It is also possible to fit a $\log(\text{strength})$ versus $\log(\text{time})$ relation to the same data without significant difference in the goodness of fit and in fact this approach gives an indistinguishable result in the final calculation).

If the rate of loss of strength of the glassfibre is directly related to the rate of some chemical reaction (e.g. at the cement/fibre interface) and

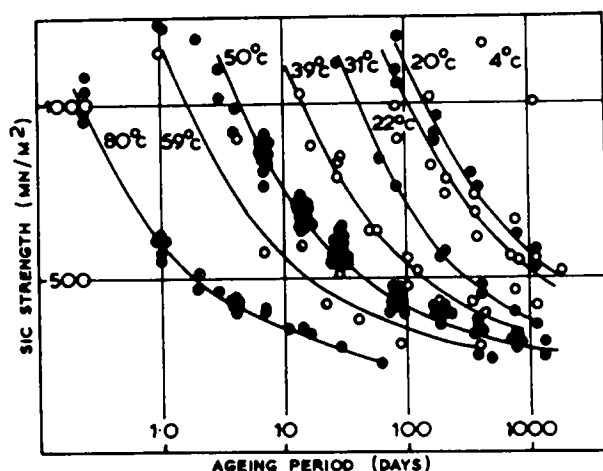


FIG.2 SIC Strength Retention in Water at Various Temperatures

further if the time taken for the SIC strength to fall to any given value (σ_{SIC}) be regarded as an inverse measure of the rate of strength loss, then an Arrhenius type relationship may be expected between the time taken for the SIC strength to fall to a given value in a particular accelerated test, and the temperature of that test;

$$\text{i.e. } \log_{10} (\text{time to reach } \sigma_{SIC}) \propto 1/T$$

where T is the absolute temperature of the accelerated test.

A family of such plots is shown in Figure 3 covering the time taken to reach strengths ranging from 1000 MN/m^2 to 300 MN/m^2 in experiments conducted at temperatures between 4°C and 80°C . The individual points were read off the best fit curves shown in Figure 2 and were restricted to the regions of Figure 2 for which actual strength measurements had been taken. This means that no extrapolation has been done at this stage, the points on the 350 MN/m^2 line in Figure 3 (for example) are restricted to temperatures above 39°C because strength had not yet fallen to that value in the lower temperature experiments and the one 4°C result only occurs on the 1000 MN/m^2 line.

In Figure 3 the data points at all strength loss levels are well fitted by a family of parallel straight lines showing good agreement with the assumed Arrhenius type relationship and also indicating a similar temperature dependence over the whole of the range of strength covered by the curves in Figure 2. We conclude from this that the form of the SIC curves in Figure 2 is the same at all temperatures and that knowledge of the complete strength loss curves obtained at high temperatures may be used to predict the values of strengths expected at very long times and lower temperatures. Interpolation on, and extrapolation of, the straight lines in Figure 3 thus provides the means of constructing SIC strength-time curves at any chosen temperature in the range 0 – 100°C and of extending the existing low temperature curves in Figure 2 to very long times.

It can be seen from the number of data points in Figure 2 that 50°C has been the most widely used acceleration temperature in the present work. It was selected because it gave results within a reasonable experimental time scale without excessive danger of altering the nature of the chemical reactions between glass and cement (3). Because the $\log(\text{time})$ – v – $1/T$ lines in Figure 3

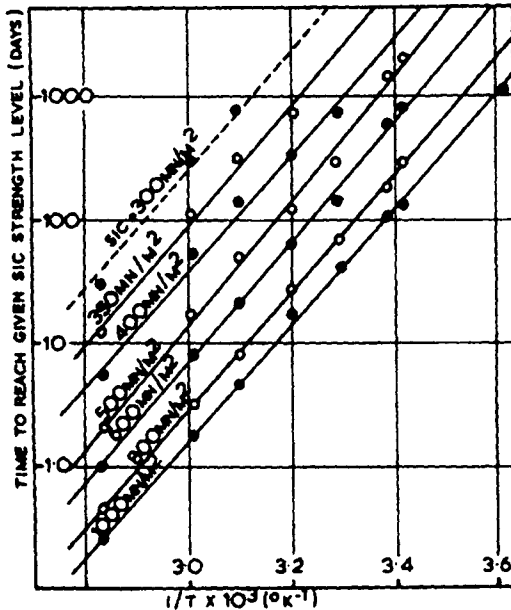


FIG.3 "Arrhenius" Plots

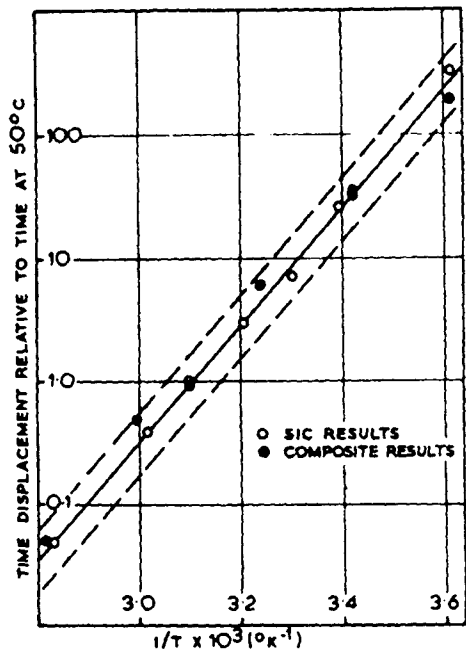


FIG.4 "Normalised Arrhenius" Plot

are all parallel to each other it is possible to "normalise" them into one single line by plotting the logarithm of the time (for a given strength loss) at some temperature T relative to the time at some standard temperature - against $1/T$. This effectively combines and averages all the data for different strength levels, given in Figure 3, into one overall picture of the relative acceleration of strength loss at different temperatures. It is also a convenient way of correlating strength changes in different accelerated test conditions and of relating them to the changes expected over very long periods of time at much lower temperatures.

This normalisation has been carried out to produce Figure 4, taking the widely used 50°C condition as the standard for comparison. The logarithm of the ratio of the time taken for the SIC strength to fall to a given value at t°C relative to the time to fall to that value at 50°C, was plotted against the inverse of the absolute temperature corresponding to t°C - the times being read off the fitted curves in Figure 2.

The good linearity of the $\log(\text{time})-v-1/T$ plot shown in Figures 3 and 4 indicates that essentially one chemical reaction or corrosion mechanism is controlling the strength changes over the whole experimental temperature range of 20°C to 80°C. This, taken with the good reproducibility of the results gives confidence in the use of accelerated tests, completed in short and convenient experimental times, to predict strength behaviour at low temperatures over longer times. Despite the earlier reservations regarding possible changes in the nature of the reactions involved, it now appears reasonable to use higher temperatures than 50°C for these accelerated tests and thus obtain information even more quickly.

It will be shown later on that the temperatures relevant to practical GRC use environments are in the range 0 to 25°C. A relatively small linear

extrapolation is required in Figure 4 in order to extend the given time/temperature relationship from the experimental range of 20°C to 80°C which contains the bulk of the results, to this use temperature range of 0-25°C. The results discussed above indicate that it is reasonable and valid to make this extrapolation - and therefore valid to assume that changes in glass strength in a cement environment, which will occur over many tens of years in real conditions, may be simulated over a few days or months in accelerated tests at high temperatures.

Accelerated Ageing of GRC Composites

In parallel with the early development of strand in cement tests, preliminary accelerated ageing experiments were being tried on composites. Again these involved immersion in hot water for various periods of time.

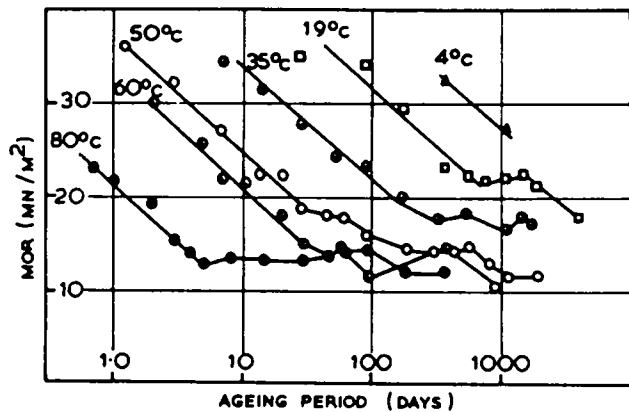
As discussed previously the strength of a fibre reinforced composite is a complex parameter depending on many factors including fibre content, orientation and distribution as well as fibre matrix bond strength and matrix strength - all of which may vary from composite to composite, and may vary with time in any one composite. Therefore experiments were concentrated on one relatively standard form of sprayed GRC which was being widely used. This contained about 5% to 6% by weight₃ of chopped Cem-FIL AR glass fibre in a composite of density about 2 gm/cm³ (i.e. 4% by volume, (10),) and was produced in the form of a flat sheet, dewatered by vacuum suction (1, 2, 8). The fibres formed an approximately random two-dimensional mat in the cement or cement/sand mortar matrix.

Early experiments had shown that the strength-time curves of composites immersed in hot water showed two distinct regions; a steadily falling initial portion followed by a constant or near constant strength region. Separate experiments had established that the rate of loss of strength in the falling region was uninfluenced by cycling between hot and cold, wet and dry, or hot wet and freezing conditions but was essentially governed by the time spent at the wet, elevated temperature condition. The initial strength fall occurred more rapidly at higher temperatures but the very long term strength appeared to be approximately independent of temperature at higher temperatures - although possibly slightly temperature dependent at lower temperatures (< 50°C).

Figure 5 shows the results of several series of tests in which strips of composite 150 mm long x 50 mm wide x 6-8 mm thickness were immersed in water at temperatures ranging from 4°C to 80°C. Groups of six samples were removed after various ageing periods and their flexural strengths measured. The points in Figure 5 represent the overall average of results of several repetitions of these experiments using different composite boards manufactured and aged on separate occasions.

From Figure 5 it can be seen that, at the higher accelerated ageing temperatures of 60°C and 80°C there is an initial fall in strength - and then a sharp transition to a constant strength region for the remainder of the ageing period. Ageing at the lower temperature of 50°C gives a similar fall in strength over the earlier portion of the curve, there is then a smoother, more gradual, transition to the same long term strength at the end of the ageing test. At even lower temperatures of 4°, 19° and 35°C the initial falling portion of the curve is still parallel to the higher temperature results - but occurs much more slowly. At 19°C and 35°C there are signs of a distinct levelling at a higher stress level than that reached in tests at 50°C to 80°C. This effect has been reported previously (4) and appears to be associated with

FIG.5 Strength Retention of GRC Composites in Water at Various Temperatures



a high degree of hydration causing void and fibre bundle filling in the composite under these conditions (see later). However subsequent results (e.g. reference 5 and the last data point on the 19°C curve of Figure 5) indicate that there is still a gradual loss of strength occurring. Thus it seems reasonable and conservative (or cautious) to assume that, over a very long period, the strengths at these lower temperatures will ultimately reach the level indicated from the constant strength regions of the higher temperature curves.

At first these test results were interpreted in a purely empirical way and accelerated ageing was carried out at temperatures of 50°C and above, until the composite strength had reached the level portion of the curves - which was regarded as a "fully aged" condition for GRC. However as information on the temperature dependence of the glass fibre strength in cement became available from the SIC tests described above, it became clear that the slope of the falling strength region - at all water immersion temperatures - was directly related to the SIC curves at the same temperatures.

Further, it was possible to predict the falling strength regions of the composite curves numerically from the SIC strand strength results using plausible values for K in the relation:-

$$\sigma_c = K \sigma_f V_f$$

where σ_c = flexural strength of composite material

σ_f = strength of glass fibre strand from SIC tests.

K = constant covering bending/tensile strength ratios, fibre orientation and fibre strength utilisation efficiency.

V_f = fibre volume content in the composite.

This will be discussed in more detail in a subsequent publication but is based on the Law of Mixtures relationship for composite strength and the knowledge that these composites undergo at least some multiple cracking (10, 11, 12) over the falling strength regions so that the cement/mortar does not contribute to the ultimate failure strength of the composite.

From this discussion it would be expected that the temperature dependence of strength loss in composites should be the same as that for fibres. This is

so and is demonstrated in Figure 4 where points showing the relative rate of strength loss for the falling strength region of the composite test results at various temperatures are seen to be closely on the line of the Arrhenius type plot obtained from the glass strand (SIC) tests.

It was concluded that:

- (a) accelerated ageing of composites over relatively short times in water at elevated temperatures could be used to produce materials in which the strength of the glass reinforcement had been reduced to a level corresponding to that attained over many years at lower temperatures.
 - (b) The temperature coefficient given in Figure 4 may be used to make quantitative estimates of particular time transpositions.
- and
- (c) The strength of the composite after such accelerated tests would be representative of its behaviour after the estimated (very long) periods at lower temperatures.

Weathering of GRC Composites in Real Climatic Conditions

It was pointed out in the introduction that the only unequivocal long term strength data for a building material must come from exposure of samples to the full vagaries of real climatic changes over very many years. Such weathering programmes, particularly if spread over a range of climates, also provide a basis for evaluating the validity of accelerated test methods and establishing the comparison between accelerated tests and real conditions. When the real weathering programmes have been running for some years - and the comparison established on the basis of reasonable experience and cross reference - then the accelerated tests may be used with confidence to extrapolate strength predictions for many years in the future in different climates. That is the position now reached with Cem-FIL reinforced GRC.

The earliest weathering programmes were established at the UK Building Research Establishment in 1968 (4) and now give results after 10 year's exposure (5). From 1972 onwards a further range of programmes were set up by Pilkington covering both different climates and some material variations (6, 7). Results from these programmes are available after exposure for 2 to 5 years. The results out to 5 years on three different sites in the UK, from many different composites, confirm the results obtained (with a few composites on only one site) from the earlier BRE tests (6).

Figure 6 shows GRC composite strengths after exposure to real weathering in comparison with strengths after the accelerated ageing tests at 50°C, 60°C and 80°C described in Section 3 above. The weathering results extend out to 10 years in the UK, five years in Toronto and two years in Bombay. The range of climate is indicated by the locations.

We can see that the form of the strength loss in real conditions is the same as that in the accelerated tests. As indicated by the early laboratory cycling tests referred to in Section 3, the changes in humidity and temperature which occur in real climates have not introduced sudden or unexpected changes in the pattern of strength behaviour.

The fall in strength in real weathering occurs much more slowly than in

the accelerated ageing tests. However there does appear to be a temperature dependence as indicated by the relative positions of the curves for the different climates, strength loss occurring most rapidly at Bombay and much more slowly at Toronto.

In fact the results show that temperature dependence in real weather is very similar to that in the accelerated tests and that the mean annual temperature of a location or climate may be used to correlate weathering strength loss with accelerated tests. This is shown in Figure 7 where rate of strength changes at a number of sites around the world have been added to the Arrhenius plot previously determined from accelerated strand and composite tests at different temperatures. Considering the very wide range of experiments and conditions the agreement with the original straight line plot is very good. There is an indication that at Bombay the rate of strength loss is significantly less than predicted - perhaps due to the concentration of rainfall into the monsoon season, followed by prolonged dry conditions in which glass/cement reactions are slowed by reduced availability of water (13). There is a similar effect with preliminary results (not shown in Figure 7) from a very dry site in Arizona. Generally however it appears possible to make a reasonable prediction of the strength loss behaviour in a variety of climates from a knowledge of the mean annual temperature and accelerated composite strength data.

The above discussion and good correlations relate to the slope and location of the initial falling strength portions of the curves. It will be noted that neither the Bombay results out to two years nor the UK results to ten years show the temporary levelling at about 22 MN/m^2 observed in the lower temperature total water immersion results at 19°C . It was suggested in Section 3 above that this difference in behaviour between high and low temperature water immersion might be due to differences in the degree and nature of cement hydration. There is some indication from microscopic studies that the morphology of hydration seen in the higher temperature accelerated water immersion experiments is more akin to that observed in weathering exposure. This is illustrated in Fig.8, there is voidage and porosity behind the fibre/cement interface after both 10 years weathering and 5 days in water at 80°C , whereas after 6 years in water at $18/20^\circ\text{C}$ the region behind the interface is generally quite a solid layer. Thus both the pattern of strength behaviour and the microscopic evidence lend support to the use of the (more conservative) higher temperature test results, shown in Figure 6, as an accelerated ageing procedure for natural weathering exposure.

The weathering sites referred to in Figure 7 cover a wide range of climatic and practical use conditions for construction materials. The mean annual temperatures (13) range from about 6°C to 28°C - mean

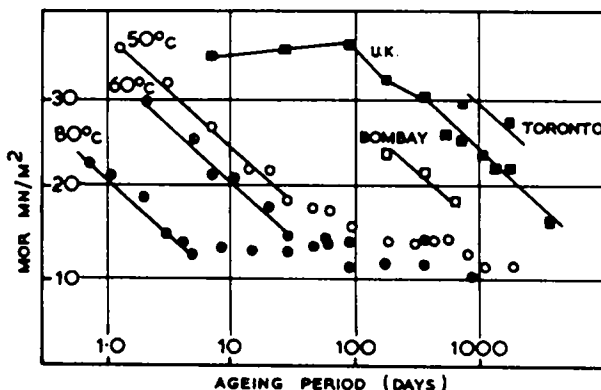


FIG.6 Strength Retention of GRC Composites in Water and Weathering

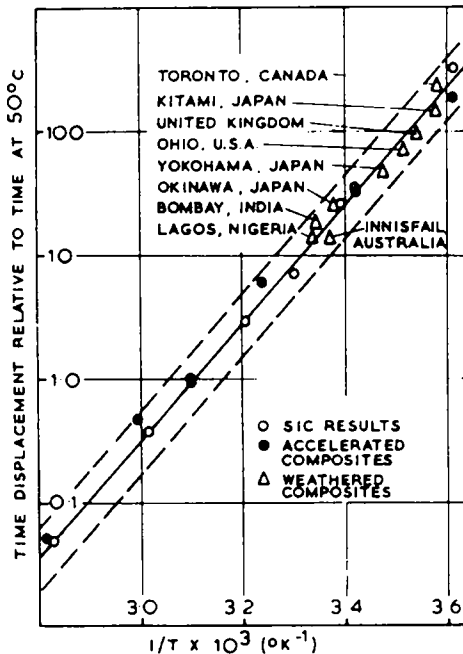


FIG.7 "Normalised Arrhenius" Plot - Including Weathering Data

annual temperatures in a temperate climate such as the UK are about 10°C. The long term strength behaviour of spray-dewatered GRC containing about

4% by volume of Cem-FIL AR glass fibre, predicted from accelerated composite strength data and the temperature coefficient of Figure 7, is plotted in Figure 9 for a cool temperate climate (M.A.T. 10°C) and a tropical climate (M.A.T. 25°C). It will be seen that strength is expected to fall slowly over some 30 to 40 years in the cooler climate - but will fall more rapidly in the hot climate. The very long term strengths in both climates will be similar, the overall average value of the accelerated composite results for times equivalent to 40 to 600 years at 10°C being 13.6 MN/m² with a standard deviation of 1.6 MN/m² from tests at temperatures of 50°C, 60°C and 80°C.

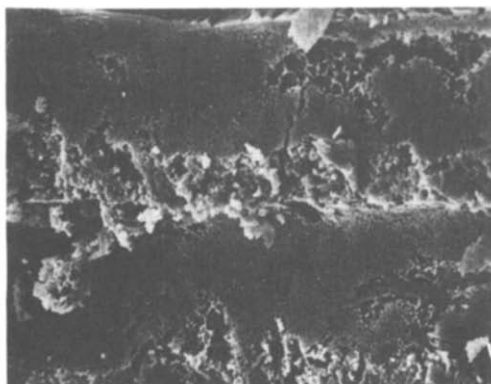
Summary and Conclusions

A simple test has been devised to measure the strength retention of glass fibre strands in a cement environment. This has proved to be extremely valuable in separating out glass fibre strength changes from the complexities of composite behaviour.

Strand in Cement strength tests with Cem-FIL AR Glassfibres were conducted over a range of temperatures from 20°C to 80°C and led to the conclusion that one chemical reaction was rate controlling over that temperature range.

Accelerated ageing tests on GRC composites immersed in water at temperatures between 50°C and 80°C showed a pattern of early strength loss followed by a constant, or near constant, strength region. The loss in strength was governed by the fibre strength changes and the temperature dependence of composite strength changes was the same as that for the glassfibre strand strengths.

The strength behaviour of Cem-FIL glassfibre reinforced composites in accelerated tests has been compared with their behaviour in a range of climates

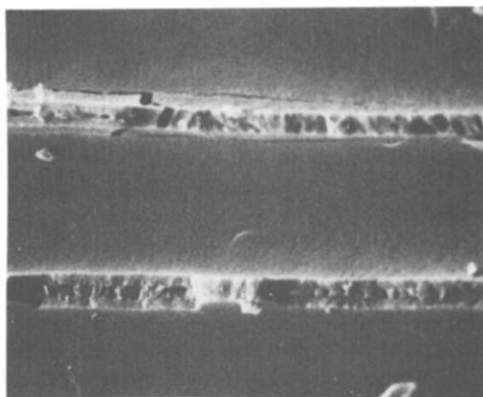


(a) After 10 years UK weather



(b) After 5 days in water at 80°C

FIG.8 Fibre-cement Interfaces

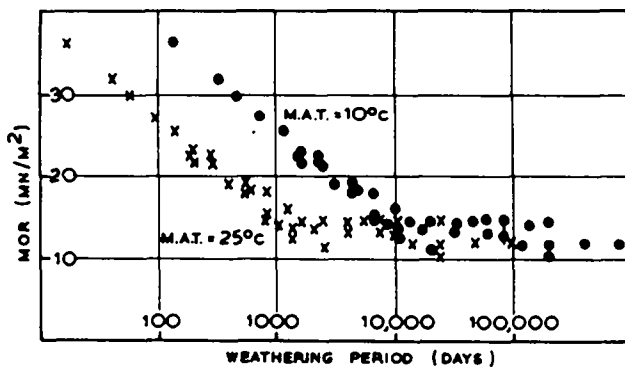
(————— = 10 μ m)

(c) After 6 years in water at 18/20°C

over periods of several years. No unusual features were introduced into the strength changes by the relatively rapid variations in conditions (e.g. temperature, humidity, rainfall) of real climates and there was a similar temperature dependence which could be related to the mean annual temperature of the weathering site.

On the basis of the established comparison between behaviour in real weather and accelerated tests, predictions have been made of composite strengths for some 100+ years in different climates.

FIG.9 Strength Retention of Cem-FIL GRC Predicted from Accelerated Test Data



Acknowledgements

The authors are indebted to Dr. E. Cohen of Owens Corning Fibreglass Ltd and Dr. K. Mishima of Asahi Glass Company for permission to use their unpublished results in Figure 7. We are also grateful to many of our colleagues at Pilkington Brothers Ltd, particular Mr. D. Ward and Mr. A. J. Aindow for making available to us further previously unpublished results. Finally we would like to record our appreciation to Dr. Majumdar and colleagues at the Building Research Establishment for much helpful discussion and for providing the sample used for illustration Fig.8(c).

This paper is published with the permission of the Directors of Pilkington Brothers Ltd and Mr. A. S. Robinson, Director of Group Research and Development.

References

1. Majumdar A. J. and Ryder J. F., Glass Technology (3), Vol.9, 1968, pp.78-84.
2. Majumdar A. J., Proc. Roy. Soc., London, Vol.A319, 1970, pp.69-78.
3. Proctor B. A. and Yale B., Phil. Trans. Soc., London, A294, 1980, pp.427-436.
4. Building Research Establishment Current Paper CP38/76, "A Study of the Properties of Cem-FIL/OPC Composites," 1976.
5. Building Research Establishment Information Paper IP36/79, "Properties of GRC:ten year results," 1979.
6. Proctor B. A., "Properties and Performance of GRC," Concrete International 1980, The Concrete Society, Proceedings Fibrous Concrete Symposium, pp.69-86.
7. Proctor B. A., "Fibre Reinforcement of Cement and Concrete," Fourth South African Building Research Congress, Capetown, May, 1979, Paper 4/3.
8. Blackman L. C. F, Proctor B. A., Smith J. W. and Taylor J. W., The Chartered Mechanical Engineer, January, 1977, pp.45-51.
9. Cohen E. B. and Diamond S., RILEM Symposium 1975, "Fibre Reinforced Cement and Concrete," edited by A. Neville (The Construction Press Ltd) pp.315-325.
10. Proctor B. A., Composites, January 1978, pp.43-48.
11. Oakley D. R. and Proctor B. A., RILEM Symposium 1975, "Fibre Reinforced Cement and Concrete," edited A. Neville (The Construction Press Ltd) pp.347-359.
12. Proctor B. A., Oakley D. R. and Wiechers W., "Composites - Standards, Testing and Design, IPC Science and Technology Press 1974, pp.106-107.
13. Tables of Temperature, Relative Humidity and Precipitation for the World, HMSO, London, 1967.